

William James on the Unity of Science

Question: Can you find a quote from William James that talks about sciences and whether they converge on a single science, or whether they deal with separate realities?

What Matters? (Flach & Voorhorst) cites William James directly on this point:

"Most thinkers have a faith that at bottom there is but one Science of all things, and that until all is known, no one thing can be completely known." — William James, *Psychology* (Henry Holt and Company), p. 1

Follow-up: Do you have access to the more complete quotation in the original source?

Yes. The citation is accurate but trimmed. It comes from the opening of Chapter I ("Introductory") of James's *Psychology: Briefer Course* (1892, Henry Holt & Co.) — an abridgment of his larger *Principles of Psychology*. The "1909" date in the *What Matters?* citation is a later Henry Holt reprint of this same text; the passage and page 1 match exactly.

The fuller passage:

"Psychology is to be treated as a natural science in this book. This requires a word of commentary. Most thinkers have a faith that at bottom there is but one Science of all things, and that until all is known, no one thing can be completely known. Such a science, if realized, would be Philosophy. Meanwhile it is far from being realized; and instead of it, we have a lot of beginnings of knowledge made in different places, and kept separate from each other merely for practical convenience' sake, until with later growth they may run into one body of Truth. These provisional beginnings of learning we call 'the Sciences' in the plural. In order not to be unwieldy, every such science has to stick to its own arbitrarily-selected problems, and to ignore all others."

James then runs through examples of exactly this arbitrary narrowing — physics assumes matter and force without justifying them, chemistry uncritically takes physics's data, physiology takes chemistry's — and lands the point on psychology itself:

"Psychology as a natural science deals with things in the same partial and provisional way... The full truth about states of mind cannot be known until both Theory of Knowledge and Rational Psychology have said their say. Meanwhile an immense amount of provisional truth about them can be got together, which will work in with the larger truth and be interpreted by it when the proper time arrives."

What This Adds

The fuller context sharpens what *What Matters?* was doing with the one-liner. James isn't just musing that all sciences *might* converge — he is explaining, as a methodological justification right at the start of a psychology textbook, why he is deliberately treating psychology as one of the provisional, arbitrarily-bounded "Sciences" (plural) rather than waiting for the unified "Philosophy" that would eventually need it. The separateness of the sciences is candidly a practical, temporary expedient, not a claim about separate underlying realities.

Why Metaphysics Matters

Question: In the same book that James talks about the unity of science, he also talks about Metaphysics. Can you pull out his discussion of why metaphysics is important?

James returns to this exact theme in the Epilogue, "Psychology and Philosophy," which opens with a section titled "What the Word Metaphysics Means" — and he explicitly cross-references his own page 1:

"In the last chapter we handed the question of free-will over to 'metaphysics.'... [T]he present writer... has no hesitation in regarding [ethics's] claim as the stronger, and in assuming that our wills are 'free.' For him, then, the deterministic assumption of psychology is merely provisional and methodological. ... [A]ll these special sciences, marked off for convenience from the remaining body of truth (cf. p. 1), must hold their assumptions and results subject to revision in the light of each others' needs. The forum where they hold discussion is called metaphysics."

Then his definition, and the argument for why it matters:

"Metaphysics means only an unusually obstinate attempt to think clearly and consistently. The special sciences all deal with data that are full of obscurity and contradiction; but from the point of view of their limited purposes these defects may be overlooked. Hence the disparaging use of the name metaphysics which is so common. To a man with a limited purpose any discussion that is over-subtle for that purpose is branded as 'metaphysical.' A geologist's purposes fall short of understanding Time itself. A mechanist need not know how action and reaction are possible at all. A psychologist has enough to do without asking how both he and the mind which he studies are able to take cognizance of the same outer world. But it is obvious that problems irrelevant from one standpoint may be essential from another. And as soon as one's purpose is the attainment of the maximum of possible insight into the world as a whole, the metaphysical puzzles become the most urgent ones of all."

The argument, condensed: every specialized science survives by quietly declining to examine its own foundational terms — geology doesn't ask what Time itself is, mechanics doesn't ask how action and reaction are possible at all, psychology doesn't ask how mind and brain "correspond." Calling a question "metaphysical" is usually just a polite way of saying it's too deep for the job at hand. But that dismissal is purpose-relative, not a verdict on the question's importance — the moment your purpose widens from a science's narrow job to "the maximum of possible insight into the world as a whole," exactly those bracketed-off questions become the most urgent ones there are. Metaphysics, on his account, isn't mystification opposed to clear thinking — it's what clear thinking turns into once you refuse to let any question get walled off by convenience.

The "(cf. p. 1)" is James pointing back at his own earlier passage on the plural, provisionally-separated sciences: the two quotes above aren't just thematically related, he wrote them as a single argument spanning the whole book, with metaphysics named as the "forum" where the separate sciences' provisional assumptions get checked against each other.

Source

William James, *Psychology: Briefer Course* (New York: Henry Holt and Company, 1892), Chapter I ("Introductory," p. 1) and Epilogue ("Psychology and Philosophy," pp. 461–462). Public domain text: [Project Gutenberg eBook #55262](#)